Math 248: Worksheet: Review

Your Name Here

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Worksheet: Comparing Statistical Models - A Student Practice

Objective: To learn how to choose the appropriate statistical model for different scenarios and perform the analysis using R.

Instructions: For each scenario, answer the questions to determine the correct statistical model. Then, complete the R code to perform the analysis and interpret the results.

Part 1: Scenario 1 - Single Mean, Paired Data, or Two-Independent Samples?

Scenario: We measure the height of 50 students at the beginning of the school year and again at the end of the year. We want to determine if there's a statistically significant change in average height over the year.

Guiding Questions:

1. **Response Variable:** What is the variable being measured?
2. **Explanatory Variable:** Is there an explanatory variable influencing the response variable? If so, what is it?
3. How many groups of data do we have?
4. Are the measurements made on the same individuals or different individuals?
5. Are we comparing the means of two independent groups or assessing a change within a single group?
6. What model is appropriate? (Single-mean, paired data, or two-independent samples?)

R Code

Generate data

```r
set.seed(123)
n_students <- 50
height_beginning <- rnorm(n_students, mean = 150, sd = 10)
height_end <- height_beginning + rnorm(n_students, mean = 2, sd = 3) # Simulate growth

# Create data frame
paired_data <- data.frame(
  StudentID = 1:n_students,
  HeightBeginning = height_beginning,
  HeightEnd = height_end
)
```

```r
# Calculate the difference in heights
paired_data$HeightDifference <- paired_data$HeightEnd - paired_data$HeightBeginning
```

7. Perform the appropriate test

```
# Paired t-test
t_test_paired <- t.test(paired_data$HeightBeginning, paired_data$HeightEnd,
                        paired = TRUE)
print(t_test_paired)
```

```
##
##  Paired t-test
##
## data:  paired_data$HeightBeginning and paired_data$HeightEnd
## t = -6.3497, df = 49, p-value = 6.749e-08
## alternative hypothesis: true mean difference is not equal to 0
## 95 percent confidence interval:
##  -3.211201 -1.667249
## sample estimates:
## mean difference
##       -2.439225
```

8. Check assumptions for paired t-test. Based on the assumption checks, is this test is appropriate for these data? Explain your reasoning.

```
## Normality of the differences
shapiro.test(paired_data$HeightDifference) # Shapiro-Wilk test for normality
```

```
##
##  Shapiro-Wilk normality test
##
## data:  paired_data$HeightDifference
## W = 0.99073, p-value = 0.9618
```

9. Interpret the p-value and confidence interval in the context of the scenario. What does your conclusion mean in terms of student height changes over the school year?

Part 2: Scenario 2 - Single Mean, Two-Independent Samples, or Paired Data?

Scenario: We want to compare the average weight (in kg) of adult men and adult women. We have independent samples of men and women.

Guiding Questions:

1. **Response Variable:** What is the variable being measured?
2. **Explanatory Variable:** What is the explanatory variable that might influence the response variable?
3. How many groups of data are there?
4. Are the groups independent or related?
5. What is the objective of the analysis (to compare the means of two groups or assess change within a group)?
6. What model is appropriate? (Single-mean, two-independent samples, or paired data?)

R Code

Generate Data

```
set.seed(123)
n_men <- 100
```

```r
n_women <- 100
weight_men <- rnorm(n_men, mean = 75, sd = 10)
weight_women <- rnorm(n_women, mean = 60, sd = 8)

weight_data <- data.frame(
  Weight = c(weight_men, weight_women),
  Gender = factor(rep(c("Male", "Female"), each = c(n_men, n_women)))
)
```

7. Perform two-sample t-test

```r
# Use if assumptions are met
t_test_weight <- t.test(Weight ~ Gender, data = weight_data, var.equal = TRUE)
print(t_test_weight)
```

```
##
##  Two Sample t-test
##
## data:  Weight by Gender
## t = -14.011, df = 198, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group Female and group Male is not equal to
## 95 percent confidence interval:
##  -19.12400 -14.40486
## sample estimates:
## mean in group Female   mean in group Male
##             59.13963             75.90406
```

8. Check assumptions for two-sample t-test. Based on the assumption checks, is this test is appropriate for these data? Explain your reasoning.

```r
#Normality
shapiro.test(weight_data$Weight[weight_data$Gender == "Male"])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  weight_data$Weight[weight_data$Gender == "Male"]
## W = 0.99388, p-value = 0.9349
```

```r
shapiro.test(weight_data$Weight[weight_data$Gender == "Female"])
```

```
##
##  Shapiro-Wilk normality test
##
## data:  weight_data$Weight[weight_data$Gender == "Female"]
## W = 0.97289, p-value = 0.03691
```

```r
# Homogeneity
leveneTest(Weight ~ Gender, data = weight_data)
```

```
## Levene's Test for Homogeneity of Variance (center = median)
##        Df F value  Pr(>F)
## group   1  3.0866 0.08048 .
##       198
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

9. Interpret the p-value and confidence interval. What does your conclusion mean in terms of

average weight differences between men and women?

Part 3: Scenario 3 - Single Mean, Two-Independent Samples, or Paired Data?

Scenario: A researcher wants to know if the average score on a cognitive test is significantly different than a known population average of 50. A sample of 25 participants takes the test.

Guiding Questions:

1. **Response Variable:** What is the variable being measured?
2. **Explanatory Variable:** Is there an explanatory variable?
3. How many groups of data are there?
4. Are the groups independent or related?
5. What is the objective of the analysis (to compare the means of two groups or assess change within a group, or compare to a known value)?
6. What model is appropriate? (Single-mean, two-independent samples, or paired data?)

R Code

Create the dataset

```
set.seed(123)
n_participants <- 25
test_scores <- rnorm(n_participants, mean = 55, sd = 5)
```

7. Perform a one-sample t-test to determine if the mean of the test_scores is significantly different from 50.

```
t.test(test_scores, mu = 50) # One-sample t-test
```

```
##
##  One Sample t-test
##
## data:  test_scores
## t = 5.1053, df = 24, p-value = 3.184e-05
## alternative hypothesis: true mean is not equal to 50
## 95 percent confidence interval:
##  52.87939 56.78731
## sample estimates:
## mean of x
##  54.83335
```

8. Check assumption of normality for the one-sample t-test. Based on your assumption check, is a one-sample t-test appropriate here? Explain your reasoning.

```
# Normality
shapiro.test(test_scores)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  test_scores
## W = 0.96745, p-value = 0.5812
```

9. Interpret the p-value. What does your conclusion mean in terms of the average cognitive test score compared to the population average?